

Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name - (Code):

Program and Semester: B Tech / MBA Tech – Computer science, IT, BTI Computer Engineering, Cyber Security ,Data science	Pre-requisite Course :					
Academic Year: 2026-27	Credit Details:	L	T	P	C	H
		2	0	2	3	4
Name of Course Faculty: Dharmesh Rathod Program: BTech IT Contact Details: 9765828205 dharmesh.rathod@nmims.edu Office Hours:	Faculty associated with the course: - 1. Prof. Dharmesh Rathod					
Pre-Course Activity: Using GenAI, Explore the following: 1. Role of Software / Need of Software 2. Categories of Software 3. What is Modeling in software making?						
Course link:						

1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance: software engineering course equips students with the foundational knowledge, practical skills, problem-solving abilities, and professional ethics necessary to thrive in the field of software development. It prepares them to tackle real-world challenges, contribute to innovative solutions, and build successful careers in the technology industry.

1.1.2 Industry Relevance: industry relevance of software engineering courses lies in their ability to prepare students with the skills, knowledge, and experiences needed to thrive in the competitive and dynamic field of software development. By staying current with industry practices and technologies, these courses ensure that graduates are ready to make immediate contributions to their organizations and continue to grow throughout their careers.

1.2 Objectives of the Course:

The principal objective of this course is to understand the basic concepts and methodologies for software engineering and to develop a foundation that can be used as the basis for further study and research in this field

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

CO-1	Explain the characteristics of various process models used in the development of a Software project
CO-2	Demonstrate an understanding of various Analysis and Design models that provide a basis for the software development
CO-3	Apply UML concepts for modeling software functionality for a given scenario
CO-4	Create test cases for validating the working of the software developed

2.2 Program Outcomes(PO) the course contributed to:

P O - 1	PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.
P O - 2	PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

P 0 - 3	PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
P 0 - 4	PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusion.
P 0 - 5	PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
P 0 - 6	PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
P 0 - 7	PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
P 0 - 8	PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
P 0 - 9	PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
P 0 - 1 0	PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
P 0 - 1 1	PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i)Independent and life-long learning ii) Adaptability to new and emerging technologies and iii) Critical thinking in the broadest context of technological change.

2.3 Program Specific Outcomes (PSO):

2.3.1 PSO-1: Demonstrate an ability to visualize, architect and create appropriate solutions for IT related projects.

2.3.2 PSO-2: Demonstrate an ability to professionally manage, monitor and safeguard IT resources.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	2	-			-	-	1	3	2
CO2	3	3	3	-	2	-			-	-	1	3	2
CO3	3	3	-	2	3	-			-	-	1	3	3
CO4	3	3	-	3	3	-			-	-	1	3	3

2.4 Student Outcomes (SO) (For ABET accredited Programs):

- 2.4.1 SO-1:** Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
- 2.4.2 SO-2:** Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
- 2.4.3 SO-3:** Communicate effectively in a variety of professional contexts.
- 2.4.4 SO-4:** Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
- 2.4.5 SO-5:** Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
- 2.4.6 SO-6:** Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.

3. Teaching-learning methodology**3.1 Instruction Plan**

Lecture No.	Topic	Teaching Method *	Blooms Level	Resources	COs mapped	Assessment and Evaluation
Unit 1: Introduction to problem solving skills						
1	Importance of Software Engineering Discussion on prerequisites, Introduction to Software Engineering including discussion of importance of Software Engineering	PPT, Whiteboard	UNDERSTAND	Standard text book & Reference book	CO1	Lab Submission

2	Importance of Software Engineering Discussion on various roles of software, Categories of Software, Understanding the Legacy Software using case study, discussion on software myths		UNDERSTAND	Standard text book & Reference book	CO1	-
3	Prescriptive Process Models Discussion on roles of process models		UNDERSTAND	Standard text book & Reference book	CO1	-
4	Prescriptive Process Models Understanding of process models via case studies		APPLY	Standard text book & Reference book	CO1	Lab Submission and Class Test-1 and TEE
5	Prescriptive Process Models Comparison of different process models, Discussion on Capability Maturity Model Integration	PPT, Whiteboard	APPLY	Standard text book & Reference book	CO2	Lab Submission and Class Test-1 and TEE
6	Agile Process Models - Understanding the need of agile methodology - Understanding the difference between agile methodology and prescriptive process models		APPLY	Standard text book & Reference book	CO2	

	Discussion on Agile Process Models- Agility, Agile Process					
7	Agile Process Models Adaptive software Development in Agile Environment		APPLY	Standard text book & Reference book	CO2	
8	Agile Process Models Extreme Programing (XP) in Agile environemnt		APPLY	Standard text book & Reference book	CO2	
9	Agile Process Models SCRUM Framework in Software Engineering		APPLY	Standard text book & Reference book	CO2	
10	UML Modeling Visual modeling with UML, Use case model		APPLY	Standard text book & Reference book	CO3	
11	UML Modeling Modeling with classes, Identifying classes and objects of real world problems in domain of Software Engineering.		APPLY	Standard text book & Reference book	CO3	Lab Submission and Class Test-1 and TEE
12	UML Modeling Defining events and attributes, process of creating class diagram.		APPLY	Standard text book & Reference book	CO3	
13	UML Modeling	PPT, Whiteboard	APPLY	Standard text book &	CO3	Lab Submission and Class

	Unified Modelling Language- Drawing State diagram			Reference book		Test-2 and TEE
14	UML Modeling Unified Modelling Language- Drawing Activity diagram		APPLY	Standard text book & Reference book	CO3	-
15	UML Modeling Unified Modelling Language- Identifying Modeling interaction and behavior		APPLY	Standard text book & Reference book	CO3	Lab Submission and Class
16	UML Modeling Unified Modelling Language- Drawing Sequence Diagram		APPLY	Standard text book & Reference book	CO3	Test-2 and TEE
17	Requirement Analysis & Design Requirement Engineering tasks, Elements of Analysis Model		APPLY	Standard text book & Reference book	CO2	Lab Submission and Class Test-2 and TEE
18	Requirement Analysis & Design Data Modeling Concepts and application for case study	PPT, Whiteboard	APPLY	Standard text book & Reference book	CO2	Lab Submission and Class Test-2 and TEE
19	Requirement Analysis & Design Discussion on Data Flow Model development for case study Discussion on Control Flow Model development for case study	PPT, Whiteboard	UNDERSTAND	Standard text book & Reference books	CO2	Lab Submission and Class Test-2 and TEE

20	Architectural Design Software Architecture, Data Design, Architectural Styles of Software		UNDERSTAND	Standard text book & Reference book	CO2	-
21	Class Test 2	PPT, Whiteboard	Apply		CO2	Lab Submission and Class Test-2 and TEE
22	Architectural Design Mapping Data Flow into a Software Architecture		UNDERSTAND	Standard text book & Reference book	CO2	Lab Submissions and TEE
23	User Interface Design Golden Rules for User Interface Design in software engineering		UNDERSTAND	Standard text book & Reference book	CO2	
24	Testing Strategies & Software Quality Test Strategies for Software, Verification & Validation Testing	PPT, Whiteboard	UNDERSTAND	Standard text book & Reference book	CO4	-
25	Testing Strategies & Software Quality Unit Testing, Integration Testing, System Testing		APPLY		CO4	Lab Submissions and TEE
26	Testing Strategies & Software Quality SQA Activities Capability Maturity Model Integration(CMMI)	PPT, Whiteboard	APPLY	Notes provided in class room	CO4	Lab Submissions and TEE

27	Importance of Software Engineering Discussion on prerequisites, Introduction to Software Engineering including discussion of importance of Software Engineering		APPLY	Notes provided in class room	CO4	Lab Submissions and TEE
28	Importance of Software Engineering Discussion on various roles of software, Categories of Software, Understanding the Legacy Software using case study, discussion on software myths	PPT, Whiteboard, TPS	UNDERSTAND	Notes provided in class room	CO4	Lab Submissions and TEE
29	Prescriptive Process Models Discussion on roles of process models	PPT, Whiteboard	APPLY	Notes provided in class room	CO4	Lab Submissions and TEE
30	Prescriptive Process Models Understanding of process models via case studies	PPT, Whiteboard, Flipped Classroom	UNDERSTAND	Notes provided in class room	CO4	Lab Submissions

4. Assessment and Evaluation Scheme

		Internal Continuous Assessment (ICA) (50 Marks)				
	Class Tests (20 Marks)			Term-work (30 Marks)		
Assessment Component	Class Test-1	Class Test-2	Lab Submissions	Assignments (Study Research Conference paper)	Mini project	Class Participation
Marks	10	10	10	05	10	05

4.1 Internal Continuous Assessment (ICA) – 50 marks

Class Test-1 conducted in the 6th week. It will be for 20 marks (1-hour duration). Marks obtained scaled down to 10 for ICA computation. Class Test-2 conducted in the 11th week. It will be for 20 marks (1-hour duration). Marks obtained scaled down to 10 for ICA computation. Lab submissions will have a weightage of 20 marks for ICA. Evaluation based on timely submissions of programming assignments given every week. In-class and Home assignments given will carry a weightage of 10 marks. Evaluation based on timely submission. **50% marks would be deducted for late submissions. (It is important that all assignments and lab submissions are done before the deadline given).** Challenging problems will carry a weightage of 5 marks.

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.3.3 (ICA + TEE) (100 marks) - 40% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Lab Submissions	Study of Research paper Assignments	Mini Project/Challenging Problems	TEE
CO-1	Y		Y	Y	Y	Y
CO-2	Y		Y	Y	Y	Y
CO-3		Y	Y	Y	Y	Y
CO-4			Y	Y	Y	Y

Laboratory details

The following 10 programming exercises will form the submission for laboratory coursework. Each programming exercise will contain 3 to 5 programs.

Sr. No.	Week #	Title	Prerequisite*	CO#	Unit
1	1-2	Conduct the literature survey to identify the case study and possible nature of problem. Shortlist at least five case study and shortlist one of them. Identify the scope of the problem selected. Also perform the following tasks. 1. Identify the end user of the system 2. Identify the functional requirements of the project 3. Identify the non-functional requirements of the project 4. Identify the feasibility of the project 5. Finalize one problem out of five selected 6. Frame the final problem statement. 7. List down the list of user requirements, system requirements.	Software Development Life Cycle / Literature survey	CO1	Unit-1

		8. Identify the ambiguities, inconsistencies, incompleteness from the requirements gathered. 9. List down the development plan for the selected problem statement.			
2	3-4	Select appropriate Generic Process Model/ Evolutionary process Model/ Agile Model for your project.	Software Development Life Cycle /Literature survey	CO2	Unit-1
3	5	Design level-0 Data flow diagram , Design level-1 and Level-2 Data flow diagram	Process models, concept of Requirement gathering	CO2	Unit-2
4	6	Development of Use case diagram, Documentation of use cases. Activity Diagram	Knowledge of ER diagram, Process model, Requirements	CO2	Unit-3
5	7	Design State chart diagram using case study's Control specifications. Make use of Star UML software for design	Knowledge of ER diagram, Process model, Requirements	CO2	Unit-4
6	8	Design sequence diagram, class diagram and state transition diagram for selected problem	Process Model, DFD, Requirement Engineering	CO3	Unit-4
7	9-10	Design the appropriate user interface diagram for your project using three golden rules.	User Interface Design	CO3	Unit-5
8	11-12	Proposed <u>Task 8: Bug Reporting with Bugzilla or Mantis</u> Aim: To simulate a software QA process using a bug tracking tool. Industry Use: Bug tracking is a core activity in QA teams.			

		Instructions: - Setup or use an online Bugzilla/Mantis instance. - Create 3-5 sample bug reports (severity, steps to reproduce, expected vs actual). - Assign bugs and simulate fixing lifecycle (Open → In Progress → Resolved → Closed). <u>Task 9: Agile Sprint Execution with Burndown Chart</u> Aim: To track the team's sprint performance using burndown charts. Industry Use: Product managers use this to measure sprint success. Instructions: - Create a sprint backlog with estimated story points. - Use Jira to track task status daily. - Generate a burndown chart at end of sprint. - Analyze deviations from expected progress.			
9	13	coding for your project according to the designs analyzed in EXP.4,5	Requirement Engineering, DFD, CFD and UID design	CO4	Unit-6
10	14	To Design Test Cases for any two functionalities of your project. The test case has to be designed for the case study selected in experiment 1.	Code, understanding about unit testing, integration testing	CO4	Unit-7

1. Tutorial Plan

2. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

3. GenAI Usage

3.1 Pre-class Activity:

3.2In-Class Activity:**4. Academic Integrity Statement**

Original work expected from students for all of the **assigned assessment work**. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**